

# Lab 6 Bonus Part

## 1 Instructions

- Modify the tail source with wide swing cascode current mirror, you should use only mosfets like the one shown in the sildes.
- Use only one ideal current source
- You should make  $V_{DS_2} = 50mv + V^*$  (drive mirror device deeper into saturation)
- Change  $V_{INCM_{DC}}$  to be  $1.8-0.5-V_{GS}$  (0.5V votlage complince for cascode current mirror)
- Report Operating point and  $A_{v_{cm}}$  vs frequency

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### The Magic Battery

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- ❑ To generate voltages we use  $I$ -to- $V$ : diode connected transistor
- ❑ Assume M1-M4 have the same  $W/L$ :  $V_B > V_{TH3,4} + 2V_{ov1-4}$   
 $V_{ov5} > 2V_{ov1-4} \rightarrow L_5 > 4L_{1-4}$
- ❑ Always select  $V_B > V_{B,min}$  (e.g.,  $L_5 \sim 6L_{1-4}$ )
  - Need to drive M1 and M2 a bit deeper into saturation
  - Account for body effect of M3 and M4

